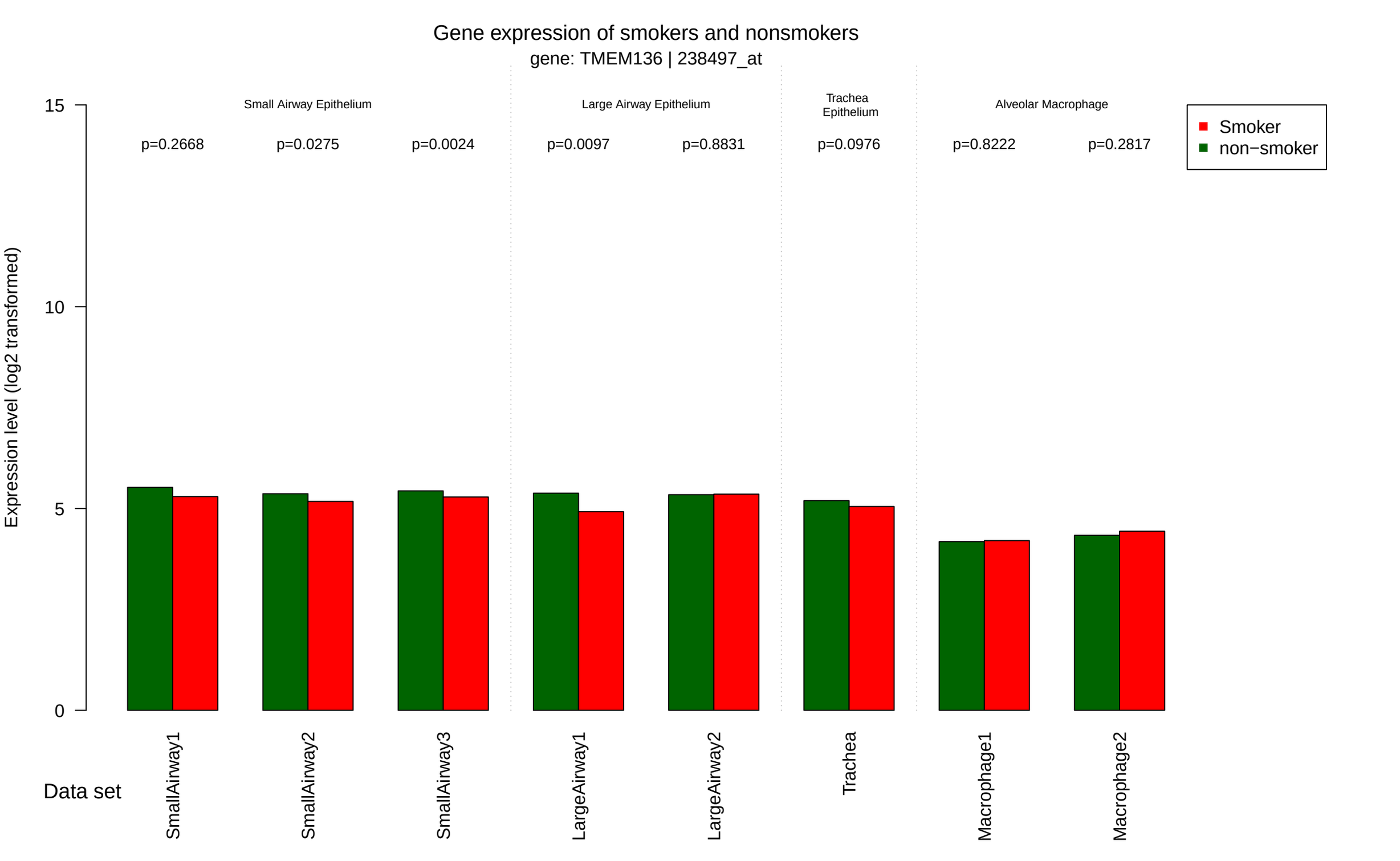
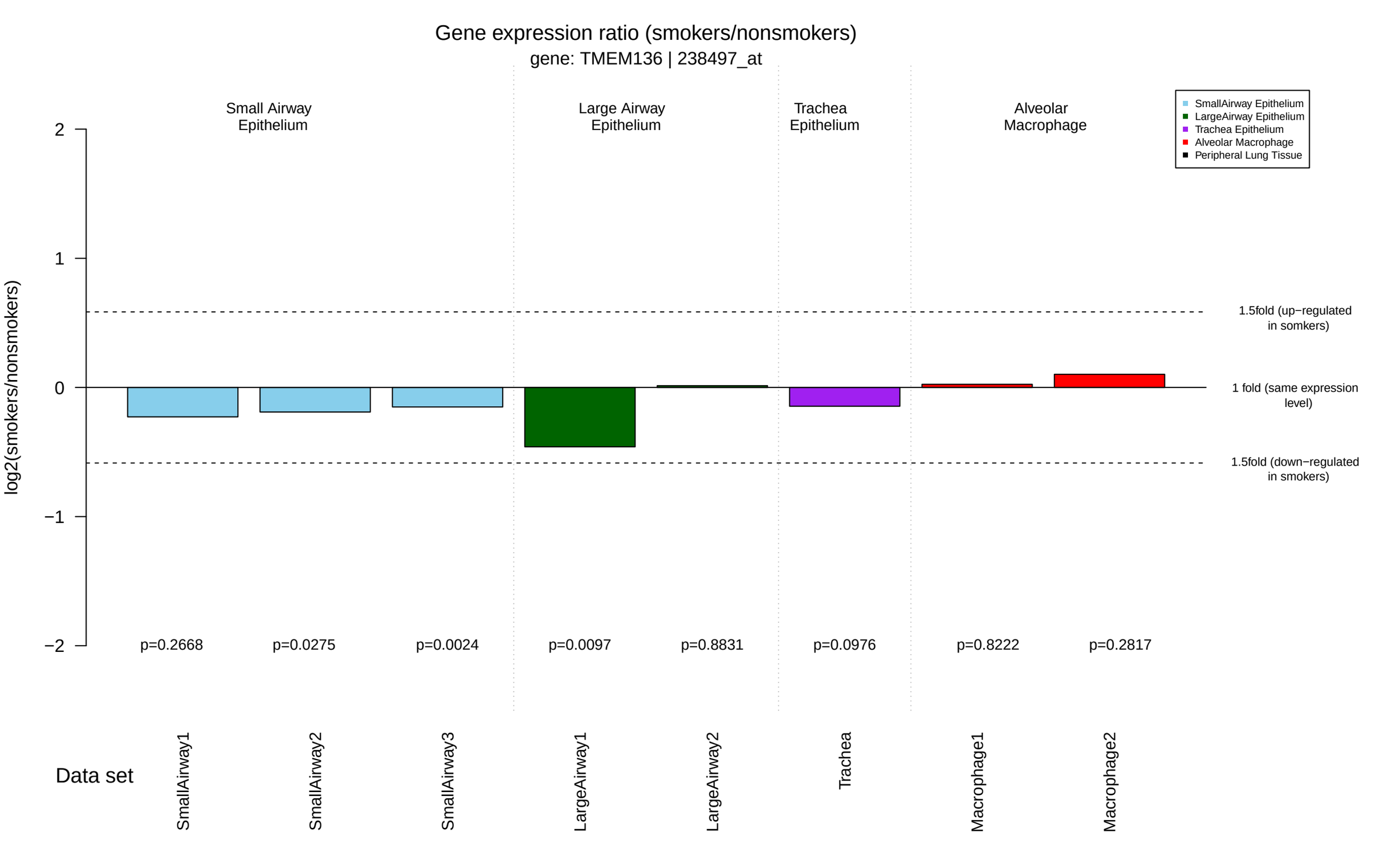
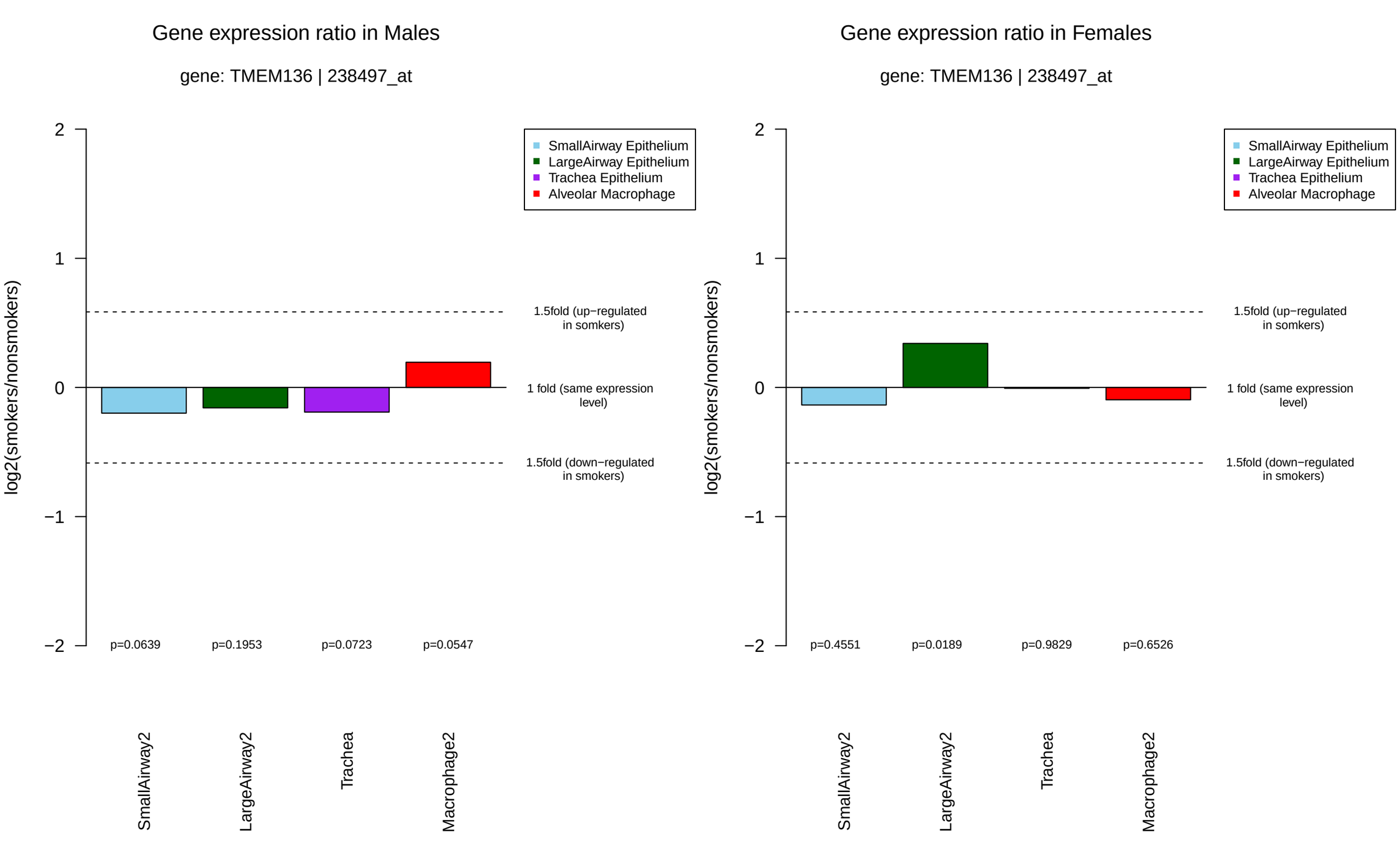
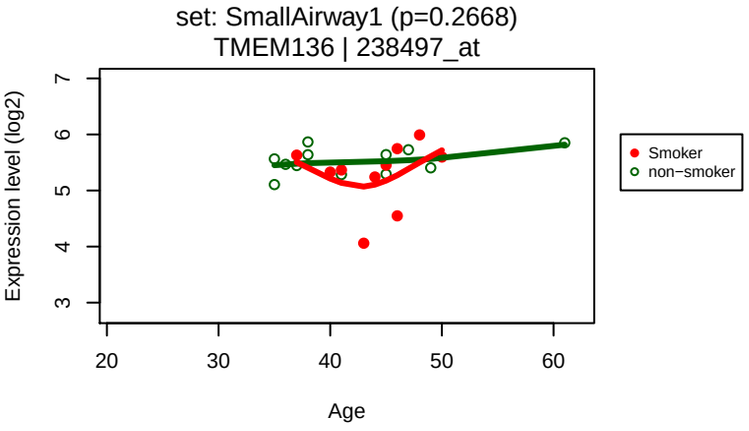
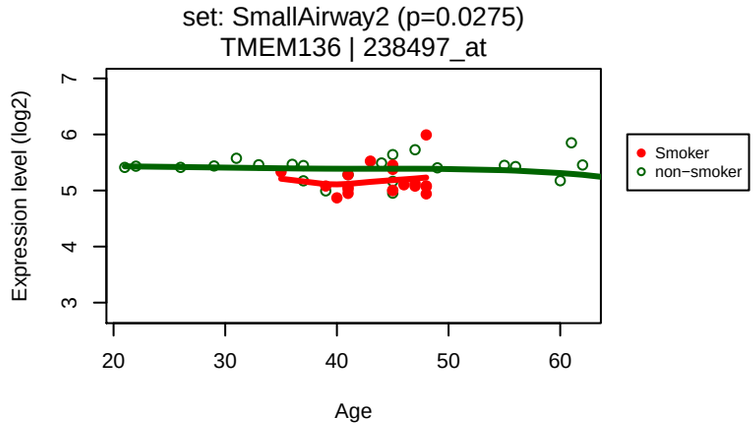
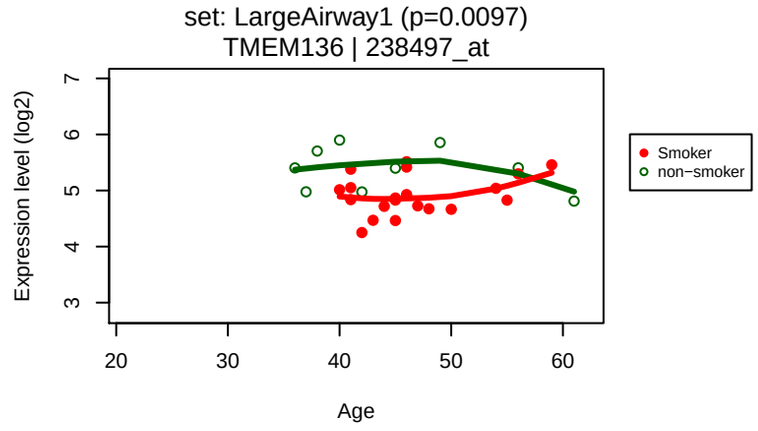

Gene expression of smokers and nonsmokers
gene: TMEM136 | 238497_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2668
p=0.0275
p=0.0024
p=0.0097
p=0.8831
p=0.0976
p=0.8222
p=0.2817
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

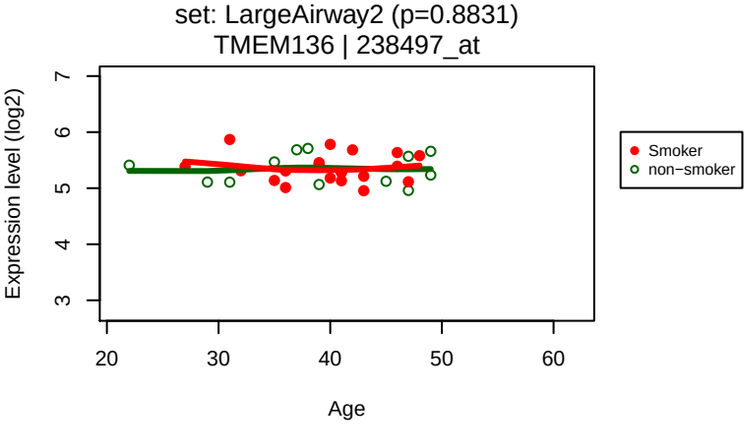

Gene expression ratio (smokers/nonsmokers)
gene: TMEM136 | 238497_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2668
p=0.0275
p=0.0024
p=0.0097
p=0.8831
p=0.0976
p=0.8222
p=0.2817
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

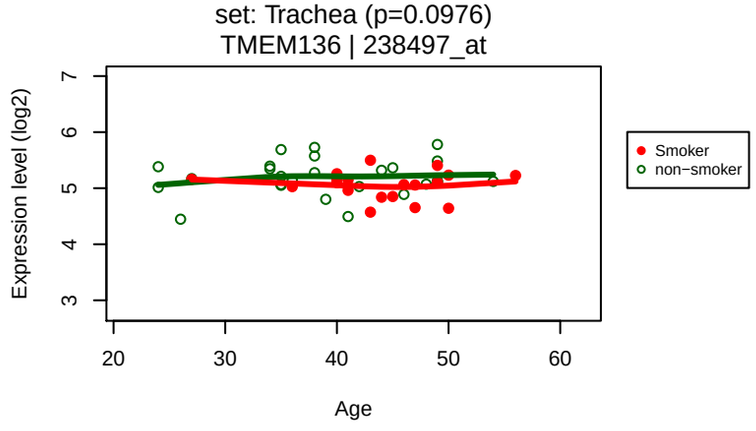

Gene expression ratio in Males
gene: TMEM136 | 238497_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0639
p=0.1953
p=0.0723
p=0.0547
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM136 | 238497_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4551
p=0.0189
p=0.9829
p=0.6526
SmallAirway2
LargeAirway2
Trachea
Macrophage2

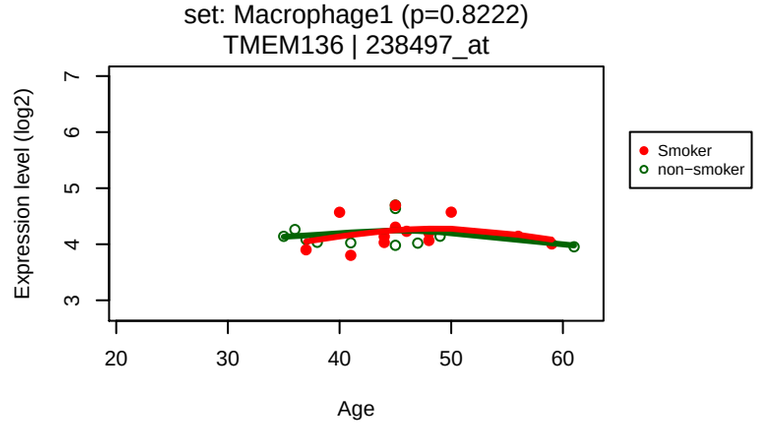

set: SmallAirway1 (p=0.2668)
TMEM136 | 238497_at
Expression level (log2)
Age
Smoker
non−smoker

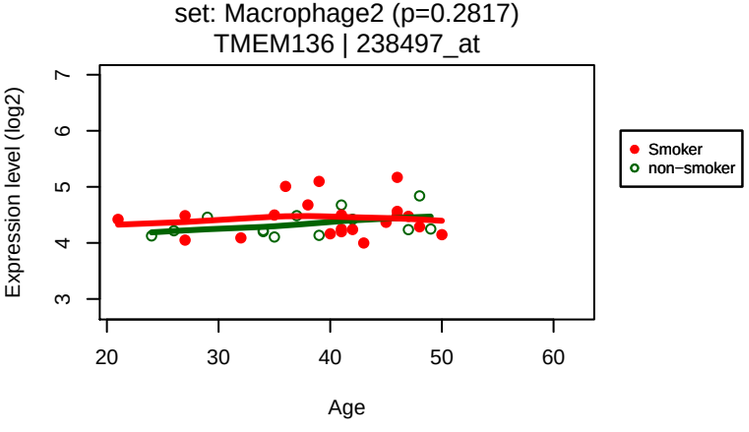

set: SmallAirway2 (p=0.0275)
TMEM136 | 238497_at
Expression level (log2)
Age
Smoker
non−smoker

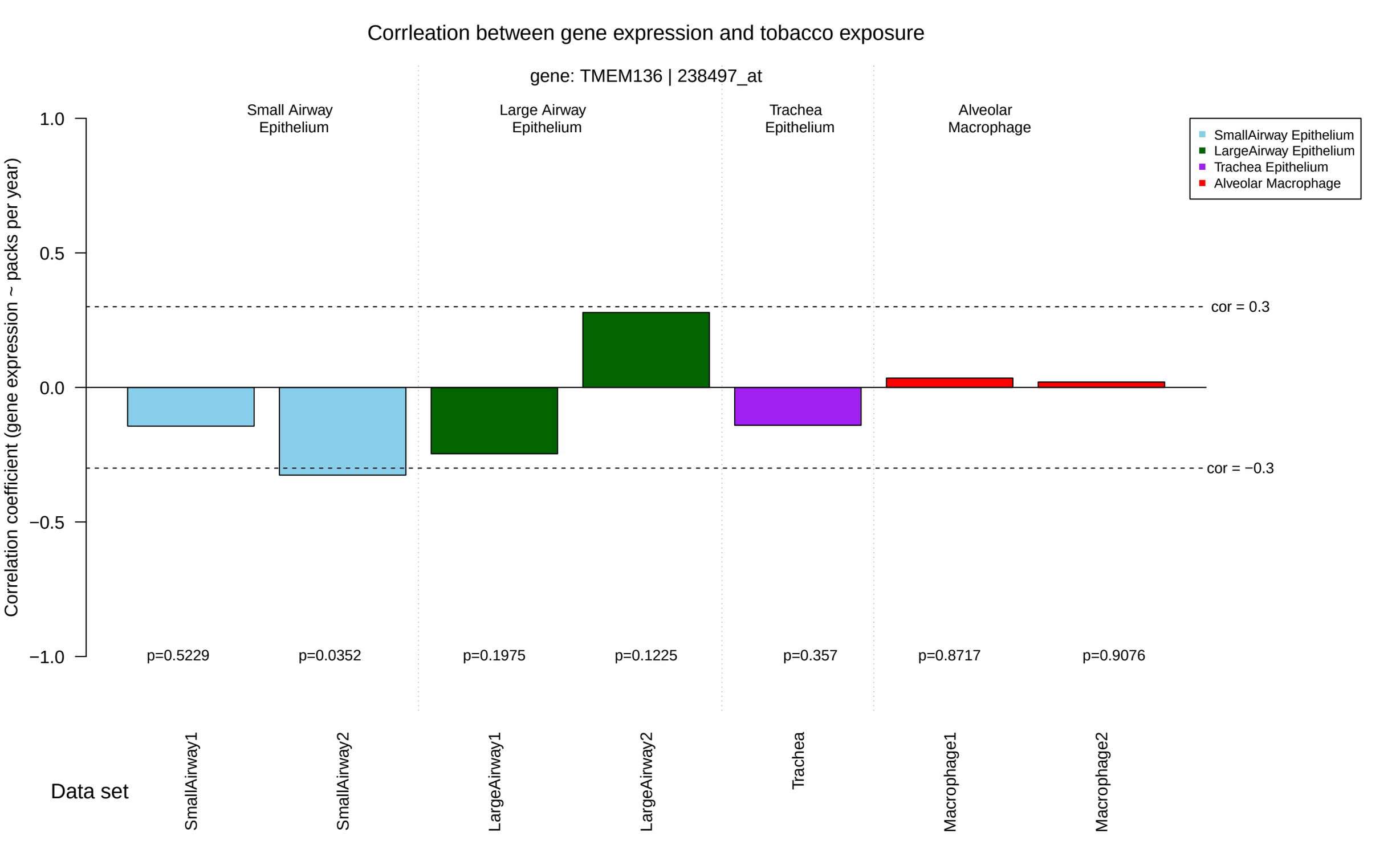

set: LargeAirway1 (p=0.0097)
TMEM136 | 238497_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8831)
TMEM136 | 238497_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0976)
TMEM136 | 238497_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8222)
TMEM136 | 238497_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2817)
TMEM136 | 238497_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM136 | 238497_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5229
p=0.0352
p=0.1975
p=0.1225
p=0.357
p=0.8717
p=0.9076
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2